

Computational Semantic Analysis of the Voynich Manuscript

An Integrated Anatolian-Islamic Medical Hypothesis and EVA-Based Semantic Resonance Framework

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Abstract

The Voynich Manuscript (Beinecke MS 408) remains one of the most extensively studied undeciphered codices in the history of cryptography, linguistics, codicology, and manuscript studies. Previous approaches have often focused on direct decipherment, cryptographic substitution, linguistic identification, or isolated statistical analysis. This paper proposes an integrated computational and historical framework that combines two complementary layers: first, a historical-contextual Anatolian-Islamic medical hypothesis based on botanical, structural, codicological, and statistical observations; second, a computational EVA-based semantic resonance framework using modern Natural Language Processing (NLP), transformer embeddings, semantic clustering, and contextual medieval medical corpora.

The historical layer examines whether the manuscript is consistent with a 15th-century copy of an earlier Anatolian-Islamic medical or pharmacological text associated with the Seljuk medical tradition. Evidence considered includes botanical identification, page-layout structure, Islamic pharmacopoeia parallels, Divriği Darüṣṣifa contextualization, Zipf-law behavior, morphological patterns, and character-frequency similarity to medieval Turkish-like profiles.

The computational layer extends this hypothesis through reproducible NLP pipelines, including EVA token processing, semantic family modeling, transformer-based embeddings, cosine-similarity analysis, PCA visualization, and unsupervised semantic clustering. Preliminary results suggest measurable contextual convergence between EVA-derived semantic structures and medieval medical-botanical domains, particularly herbal preparation, hydrotherapy, medicinal processing, bodily regulation, gynecology, and astrological medicine.

This work does not claim definitive decipherment of the Voynich Manuscript. Instead, it presents a reproducible, multi-method research framework for evaluating contextual semantic hypotheses concerning Voynichese and related undeciphered or partially understood symbolic systems.

1. Introduction

The Voynich Manuscript has resisted definitive decipherment for more than a century. It consists of an unknown script accompanied by botanical, astronomical, balneological, pharmaceutical, and cosmological imagery. Although carbon dating places the vellum in the early 15th century, the manuscript's iconographic and textual properties remain unresolved.

Foundational research by Currier, Friedman, D'Imperio, Landini, Takahashi, Stolfi, Zandbergen, and others demonstrated that Voynichese is not easily dismissed as random noise. Its token distribution, repetition patterns, internal structure, and folio-level variation suggest a system with linguistic-like regularities.

The present work proposes that the strongest research path is not immediate literal translation, but structured semantic-contextual modeling. Specifically, it integrates:

1. historical and codicological hypothesis testing;
2. botanical and pharmacological comparison;
3. EVA corpus statistics;
4. transformer-based semantic analysis;
5. medieval medical corpus comparison;
6. reproducible computational tooling.

The manuscript is treated not as an isolated cryptogram, but as a possible historical medical-text artifact whose semantic structure may become partially measurable through contextual resonance with relevant

medieval corpora.

2. Research Position and Scope

This project should be interpreted as:

- exploratory computational humanities research;
- semantic-contextual NLP experimentation;
- historical hypothesis modeling;
- medieval corpus comparison;
- reproducible framework development.

This project does **not** claim:

- definitive translation of Voynichese;
- proof of authorship;
- proof of geographic origin;
- direct symbol-to-letter equivalence;
- established linguistic decoding.

The appropriate claim is more limited but scientifically meaningful:

The framework tests whether EVA-derived structures show measurable contextual convergence with historically plausible medieval medical corpora and visual-codicological features associated with Anatolian-Islamic manuscript traditions.

3. Integrated Hypothesis

The working hypothesis is that MS 408 may be consistent with a 15th-century copy of an earlier medical-pharmacological text rooted in the Anatolian-Islamic or Seljuk medical tradition.

This hypothesis has three components:

3.1 Historical Layer

The manuscript may preserve content related to medieval Islamic pharmacopoeia, herbal medicine, hydrotherapy, astrological medicine, and bodily regulation. Institutions such as the Divriği Darüṣṣifa (1228 CE), Kayseri Darüṣṣifa, Sivas medical institutions, or related Anatolian medical centers form plausible contextual anchors, but no direct documentary attribution is claimed.

3.2 Linguistic Layer

Voynichese may encode, reflect, or transform a hybrid linguistic environment involving Anatolian Turkish, Persian medical terminology, Arabic pharmacological vocabulary, or a phoneticized transmission system. The EVA script may not map directly onto any known alphabet.

3.3 Computational Layer

Even without literal decoding, semantic regularities may be detectable through:

- lexical-family structures;
 - frequency distributions;
 - suffix/prefix behavior;
 - embedding-space proximity;
 - contextual clustering;
 - comparison with restricted medieval corpora.
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4. Prior Historical-Contextual Evidence

A prior working paper associated with this project developed a multi-method Anatolian-Islamic medical hypothesis using four main lines of evidence:

1. botanical analysis of selected herbal folios;
2. structural comparison with Islamic medical manuscripts;
3. cross-reference with Islamic pharmacological sources such as Ibn Sina and Ibn al-Baytar;

4. statistical analysis of the Takahashi EVA corpus.

The preprint argued that these lines of evidence are collectively consistent with, but do not prove, an Anatolian-Islamic medical context.

5. Botanical Analysis Layer

The botanical layer focuses on whether plant illustrations in the herbal section correspond to flora and pharmacological categories known from medieval Anatolia and Islamic medicine.

The prior analysis examined early herbal folios and proposed moderate- to high-confidence identifications for several plants, including:

Folio	Proposed Identification	Contextual Significance
f2r	Cirsium/Carduus	Islamic pharmacopoeia; thistle/cardoon-type medicinal use
f3r	Rumex sp.	Anatolian flora; documented in Ibn Sina-type traditions
f4r	Thymus/Satureja	Central Anatolian medicinal herb context
f6r	Papaver somniferum	analgesic/narcotic pharmacology
f7r	Colchicum autumnale	documented medicinal/toxic plant
f10r	Centaurea sp.	Anatolian diversity and pharmacological use
f11r	Juniperus sp.	Islamic pharmacological tradition
f13r	Mandragora officinarum	narcotic/analgesic and humoral medicine context
f15r	Hypericum perforatum	documented medicinal plant

These identifications remain probabilistic and require specialist botanical review. Their value is not as proof, but as a contextual signal: several proposed identifications are plausible within Anatolian and Islamic pharmacological traditions.

Particularly notable is the clustering of possible narcotic or analgesic plants such as Papaver, Mandragora, and Colchicum in early folios, which may be consistent with therapeutic rather than alphabetic organization.

6. Iconographic and Structural Layer

Several structural and visual conventions may align more closely with Islamic or Arabic medical manuscript traditions than with standard European herbals.

Features considered include:

- multiple text blocks distributed around central plant illustrations;
- root exaggeration emphasizing medicinally active parts;
- lateral labels or tokens near plant components;
- parallel column structures;
- serpent imagery associated with antidote/antivenin contexts;
- solar or astrological symbols associated with humoral classification.

A prior structural comparison contrasted European herbals, Arabic Dioscorides traditions, and the Voynich herbal section. The observed Voynich layout appeared closer to multi-block illustrated medical manuscript organization than to continuous European herbal text blocks.

This does not establish provenance, but supports the need to compare MS 408 with Islamic medical manuscript production traditions, especially Arabic Dioscorides and Ibn al-Baytar-related manuscript cultures.

7. Seljuk and Anatolian Medical Context

The Divriği Great Mosque and Hospital complex (1228–1229 CE) provides a historically plausible reference point for Anatolian medical culture. It should not be treated as a proven origin for MS 408, but rather as a contextual model for the type of medical, architectural, and pharmacological environment in which such knowledge could circulate.

Other plausible institutions include:

- Kayseri Darüṣṣifa;
- Sivas Darüṣṣifa;
- Konya medical institutions;
- broader Anatolian-Persian-Islamic medical networks.

Ibn al-Baytar's pharmacological tradition is also relevant because of its extensive plant-drug catalog and later circulation in Turkish-speaking Anatolian contexts.

Thus, the working hypothesis is not narrowly “Divriği produced the Voynich,” but more cautiously:

The manuscript's botanical, structural, and semantic features may be consistent with a broader Anatolian-Islamic medical knowledge environment.

8. EVA Statistical Layer

The prior working paper reported several statistical properties of the Takahashi EVA corpus:

- approximately 38,510 tokens;
- 184 folios;
- 7,380 unique token types;
- high-frequency token `daiin`;
- Zipf exponent $\alpha \approx 0.928$;
- Shannon entropy around 3.66 bits/character;
- high occurrence of `-ain` / `-aiin` -like endings;
- KL divergence suggesting closer character-frequency affinity with medieval Turkish-like profiles than with Arabic or Persian reference distributions.

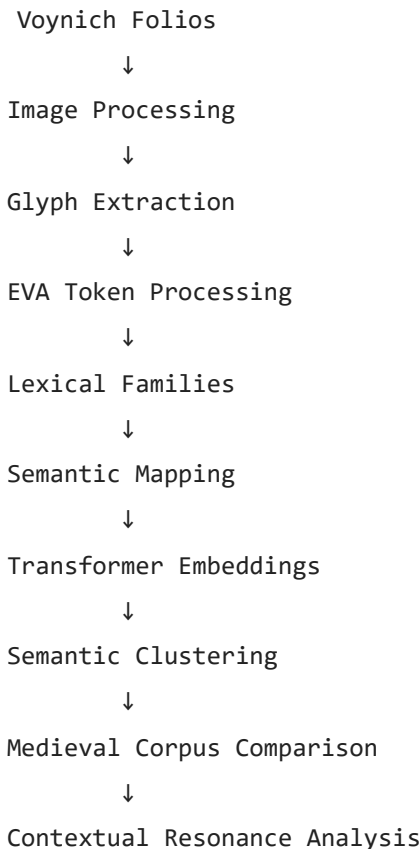
These findings should be interpreted carefully.

Zipf-like behavior supports language-like structure, but does not identify the language. KL divergence provides statistical affinity, not linguistic proof. Suffix-like patterns may suggest morphological regularity, but not direct grammatical equivalence.

Nevertheless, the statistical layer supports the broader claim that Voynichese is structured and that Turkic or hybrid Anatolian comparison is worth systematic investigation.

9. Computational Framework Architecture

The current computational framework operationalizes the above historical hypothesis using reproducible NLP and vision pipelines.



The goal is not immediate translation, but measurable semantic alignment.

10. Glyph Extraction and Visual Processing

Using OpenCV-based image processing, folio imagery was processed through:

- grayscale conversion;

- thresholding;
- contour detection;
- glyph bounding boxes;
- PNG glyph export;
- clustering of extracted glyphs.

This generated a reusable visual dataset of glyph-like units for exploratory analysis. The method is preliminary and does not yet replace expert paleographic segmentation, but provides a computational basis for large-scale visual comparison.

11. EVA Processing and Semantic Families

EVA tokens were processed through:

- tokenization;
- prefix/suffix extraction;
- frequency analysis;
- semantic family assignment;
- contextual interpretation.

Experimental EVA semantic families included:

EVA Family	Experimental Semantic Domain
qok-*	liquids / medicine / roots
yka-*	preparation / boiling / steeping
sho-*	heat / pain / solar association
pch-*	roots / cutting / mixture
dai-*	cycles / repetition / bodily rhythm
ata-*	tincture / medicinal preparation

EVA Family	Experimental Semantic Domain
qot-*	opening / channels
fac-*	action / procedure

These assignments are hypotheses used for computational modeling, not confirmed translations.

12. Medieval Corpus Construction

An experimental medieval medical corpus was created to test contextual similarity. It included fragments associated with:

- Anatolian-Seljuk medicine;
- Persian medical vocabulary;
- herbal preparation;
- hydrotherapy;
- gynecology;
- astrological medicine;
- pharmacological preparation.

Each record contained:

- title;
- date;
- language;
- region;
- topic;
- original/transliterated text;
- translation;
- keywords.

This corpus is currently small and partly synthetic/curated. Its primary purpose is methodological demonstration. A future large-scale corpus is required for stronger statistical inference.

13. TF-IDF and Transformer Embeddings

Two embedding approaches were used.

13.1 TF-IDF Similarity

TF-IDF vectorization provided a baseline lexical similarity comparison between EVA semantic structures and medieval corpus fragments.

Initial results favored medical-botanical fragments over less relevant domains.

13.2 Transformer Embeddings

Transformer-based semantic embeddings were generated using:

```
sentence-transformers/all-MiniLM-L6-v2
```

Cosine similarity produced stronger semantic convergence.

Representative results:

Corpus Fragment	Transformer Similarity Score
Seljuk Herbal Preparation Fragment	0.4290
Divriği Darüşşifa Hydrotherapy Fragment	0.4149
Persian Medical Treatise	0.4107
Seljuk Herbal Fragment	0.3967
Seljuk Women’s Medicine Fragment	0.3876

The highest-scoring fragments involve herbal preparation, hydrotherapy, pain, heat, bodily regulation, and medicinal processing.

14. Automatic Semantic Clustering

Unsupervised clustering of embedding representations produced emergent groups around:

- liquids and medicinal mixtures;
- preparation and boiling;
- heat and pain;
- cycles and bodily processes;
- herbal processing.

This suggests that the experimental semantic space is not randomly dispersed, although stronger validation against randomized baselines is still required.

15. PCA Semantic Visualization

PCA-based dimensionality reduction generated two-dimensional semantic maps showing relative positions of:

- EVA semantic structures;
- herbal preparation fragments;
- hydrotherapy texts;
- astrological medicine references;
- gynecological medical fragments.

These visualizations provide exploratory evidence of contextual proximity, but should not be interpreted as definitive proof without larger corpus validation.

16. Φ_{47} Semantic Resonance Interpretation

The broader conceptual framework can be described as semantic resonance analysis.

In this model:

symbolic structure + contextual corpus + embeddings + clustering = semantic resonance field

Rather than asking whether a token directly equals a known word, the system asks whether groups of tokens, visual features, and contextual domains resonate consistently within a historically plausible semantic space.

This approach may be applicable beyond the Voynich Manuscript to:

- undeciphered manuscripts;
 - fragmentary languages;
 - low-resource historical corpora;
 - symbolic systems;
 - partially understood scripts;
 - OCR-damaged manuscript corpora.
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17. Discussion

The integrated framework strengthens the research program in several ways.

First, the historical layer provides a plausible restricted context: Anatolian-Islamic medicine, pharmacopoeia, hydrotherapy, and manuscript culture.

Second, the computational layer makes the hypothesis testable and reproducible through scripts, data files, embeddings, clustering, and visualizations.

Third, the NLP layer avoids overclaiming direct translation while enabling measurable semantic comparison.

The strongest current convergence appears around:

- herbal preparation;
- hydrotherapy;
- medicinal processing;
- heat and pain;
- bodily cycles;
- gynecological medicine;
- astrological medical contexts.

This does not prove the manuscript's origin, but it justifies deeper investigation using larger medieval corpora.

18. Limitations

Important limitations remain:

1. The corpus is still too small.
2. Some corpus fragments are experimental or curated rather than fully sourced from large-scale primary texts.
3. Botanical identifications require expert verification.
4. Semantic families are hypotheses, not translations.
5. KL divergence does not prove language identity.
6. Transformer embeddings may reflect modern-language semantic bias.
7. No verified plaintext exists.
8. Divriği remains contextual, not documentary.
9. Baseline comparisons are not yet sufficient.
10. Statistical significance testing remains future work.

These limitations do not invalidate the framework, but they define the next research tasks.

19. Future Work

Future work should focus on building a large, real medieval semantic corpus.

Priority domains:

- medieval Anatolian Turkish;
- Persian medical texts;
- Arabic pharmacopoeia;
- Islamic medicine;
- herbals;
- hydrotherapy;
- astrological medicine;
- gynecology;
- pharmacology;
- hospital manuscripts.

The next planned project is:

Medieval Semantic Corpus Infrastructure

This will include:

- collaborative corpus uploads;
- OCR pipelines;
- metadata schemas;
- transliteration workflows;
- embeddings;
- semantic search;
- AI corpus agents;
- large-scale comparison against Voynich and other symbolic systems.

20. Conclusion

This paper presents an integrated framework combining historical hypothesis modeling and computational semantic analysis for the Voynich Manuscript.

The historical layer suggests that the manuscript may be consistent with an Anatolian-Islamic medical context, especially when botanical, structural, pharmacological, and statistical evidence are considered together.

The computational layer demonstrates that EVA-derived semantic structures can be processed through modern NLP, transformer embeddings, clustering, and corpus comparison to produce measurable contextual convergence with medieval medical-botanical domains.

No definitive decipherment is claimed.

The contribution is instead a reproducible semantic-contextual framework that can be expanded through larger medieval corpora and applied to other undeciphered or partially understood symbolic systems.

The Voynich Manuscript may therefore serve not only as an unresolved historical artifact, but also as a catalyst for the development of computational medieval semantic infrastructure.

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